EE6221 CA2 Assignment

State of the Art of General-Purpose Humanoid Robots:
From Whole-Body Control to Vision-Language-Action "Robot Brains"

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Abstract

This report reviews the current state of general-purpose humanoid robots, with a focus on one technology that decides whether they are useful: the control stack that turns camera images and a spoken instruction into coordinated motion of every joint. I compare commercial platforms (Boston Dynamics Atlas, Figure 03, Apptronik Apollo, Unitree G1) and recent research on learned whole-body control and vision-language-action (VLA) models (GR00T N1, π_0, Gemini Robotics 1.5, Helix 02, FAST, HuGe). I argue that the best current solution is a *hierarchical, learned* architecture in which a slow reasoning model, a fast visuomotor policy, and a very fast whole-body controller are trained to work together, as exemplified by Figure's Helix 02, while Atlas is the most deployment-ready hardware. I then discuss the main open challenges (data, reliability, safety certification, energy, and evaluation), likely future directions, and what I learned.

Contents

1 Introduction and scope **2**

2 Background: the layers of a humanoid control stack **2**

3 Review of the state of the art **3**

3.1 Hardware platforms 3

3.2 Learned whole-body control 3

3.3 Vision-language-action models 3

3.4 Comparison 4

4 The best solution and why **5**

5 Challenges in improving humanoid robots **5**

6 Future directions **6**

7 What I learned from this review **6**

8 Effect of this assignment on my understanding and interest **7**

1 Introduction and scope

A humanoid robot has a human-like body: two legs, a torso, two arms and, in most current designs, multi-fingered hands. The case for this form is practical rather than aesthetic. Factories, warehouses and homes were built for human bodies, so a robot with the same reach, height and hand shape can in principle use existing tools, shelves and doorways without redesigning the workplace. The same argument explains why humanoids are hard: a body with 50 or more actuated joints that must balance on two small feet is underactuated, has highly coupled dynamics, and must keep doing so while the hands interact with unknown objects.

For many years the hardware was the bottleneck. Today several companies can build reliable humanoid bodies, and the limiting factor has moved to the "brain": how the robot perceives a scene, decides what to do, and converts that decision into stable whole-body motion. This review therefore treats the overall humanoid solution but concentrates on that control stack. It connects directly to EE6221: forward and inverse kinematics of redundant arms, robot dynamics and control, vision sensors and pose estimation, sensor fusion with the Kalman filter, and vision-based control all appear as layers inside a modern humanoid.

Method. I compared systems using public technical reports, peer-reviewed or preprint papers, and official company announcements. Company announcements are treated as *claims*; I note where results lack independent verification. Each solution is judged on five criteria: (i) breadth of tasks it can perform without task-specific programming, (ii) length and reliability of autonomous operation, (iii) data efficiency and ability to transfer across robots, (iv) readiness for real deployment (hardware maturity, safety, integration), and (v) openness and quality of the evidence.

2 Background: the layers of a humanoid control stack

It is useful to describe any humanoid as a stack of loops running at very different rates.

- **L1 Actuation and state estimation** (about 1 kHz). Joint torque or current control, plus estimation of the floating base pose from IMU, joint encoders and foot contact, typically with a Kalman-type filter. The base pose is not measured directly, so this estimate is what every higher layer trusts.
- **L2 Whole-body control (WBC)** (about 50–500 Hz). Given a desired motion of the hands, feet or centre of mass, compute joint commands that achieve it while keeping balance and respecting joint and contact limits.
- **L3 Visuomotor policy** (about 10–200 Hz). Map camera images and proprioception to short-horizon targets such as hand poses, grasp commands, or a reference motion for the WBC.
- **L4 Task reasoning** (about 1–10 Hz or slower). Interpret a language instruction, break it into steps, track progress, and recover from failure.

Classical humanoids built L2 from models: zero-moment-point (ZMP) walking, model predictive control (MPC) of a simplified centroidal model, and a quadratic-program WBC that resolves the redundancy of the kinematic chain (the same null-space idea used for redundant manipulators, but with contact and balance constraints added). L3 and L4 were hand-written state machines. The common thread of the last two years is that *every layer above L1 is being replaced or augmented by learned models*, and the key design question is how those learned layers are divided and connected.

3 Review of the state of the art

3.1 Hardware platforms

Boston Dynamics Atlas (product version). Unveiled at CES on 5 January 2026, the electric Atlas is presented as a production machine rather than a research prototype [1]. The company states 56 degrees of freedom with fully rotational joints, a 50 kg payload, 2.3 m reach, water resistance and operation from −20 to 40 °C. Atlas can walk to a station and swap its own battery, so a low battery does not end a shift. It runs autonomously, by teleoperation or from a tablet, and connects to factory software (MES/WMS) through Boston Dynamics' Orbit platform, which also lets one robot's learned task be copied across a fleet. All 2026 production is committed to Hyundai's Robotics Metaplant Application Center and to Google DeepMind, which is integrating its foundation models into Atlas.

Figure 03. Figure's third-generation robot, announced in October 2025, was redesigned for volume manufacturing and for homes, with a soft textile and foam exterior, a new sensor suite, and hands with palm cameras and tactile sensing [10]. Figure has published few hard specifications; its significance in this review is as the body for the Helix 02 control model (Section 3.3).

Apptronik Apollo is a general-purpose humanoid that Google DeepMind uses as one of the embodiments of Gemini Robotics 1.5 [7].

Unitree G1 is a low-cost humanoid that has become a common research platform; for example, both HuGe [5] and MotionVLA [6] report real-robot experiments on it. Cheap, widely available hardware matters because it lets university groups test learned controllers on a real body instead of only in simulation.

3.2 Learned whole-body control

Reinforcement learning (RL) in GPU-parallel simulation, followed by sim-to-real transfer, is now the dominant way to obtain robust locomotion and whole-body motion for humanoids. The newest trend is the *behaviour foundation model* (BFM): a whole-body controller pre-trained on large motion datasets, often retargeted human motion capture, so that it learns a general repertoire of skills and can be adapted to new tasks zero-shot or with little data. Yuan et al.'s survey, published in IEEE TPAMI in 2026 [4], organises this field by pre-training approach and identifies its open problems: retraining cost for each new scenario, the gap between simulated and real dynamics, and safe adaptation.

FAST (Wang et al., February 2026) [3] is a representative recent method. It keeps a pre-trained tracking policy and adds a small *residual* policy for motions outside the training distribution, with orthogonality (Parseval) and KL constraints so that adapting to a new motion does not erase old skills. It also adds centre-of-mass observations and objectives, which explicitly brings a classical balance quantity back into a learned controller. I find this telling: the best learned controllers are not ignoring model-based insight but feeding it to the network.

3.3 Vision-language-action models

A VLA model starts from a vision-language model pre-trained on internet data and is trained further to output robot actions. The key systems are:

- π_0 (Physical Intelligence, 2024) [9] attaches a flow-matching "action expert" to a vision-language backbone so it can produce smooth, high-rate continuous actions, and trains across many robot types. It established the general recipe that later humanoid models follow.
- **GR00T N1** (NVIDIA, March 2025) [8] is an open humanoid foundation model with a *dual-system* design: a vision-language module ("System 2") interprets the scene and the

instruction, and a diffusion-transformer action module ("System 1") generates fluid motor actions. It is trained on a mixture of real robot teleoperation, human videos and synthetic data, and the weights are public.

- **Gemini Robotics 1.5** (Google DeepMind, October 2025) [7] pairs a VLA with a separate embodied-reasoning model (Gemini Robotics-ER 1.5) that plans and checks progress. The VLA can "think" in natural language before acting, and a *motion transfer* mechanism lets skills learned on one robot run on ALOHA, a bi-arm Franka and the Apollo humanoid without robot-specific post-training.
- **Helix 02** (Figure, early 2026) [2] extends Figure's original upper-body Helix model to the whole body. It adds a learned whole-body controller ("System 0") trained on over 1,000 hours of human motion data and sim-to-real RL, beneath the existing System 1 visuomotor policy and System 2 reasoning model. Figure demonstrated a four-minute, uninterrupted, fully autonomous task of unloading and reloading a dishwasher across a kitchen, which it describes as the longest and most complex autonomous humanoid task to date, and later claimed multi-hour autonomous work shifts. These are company claims without independent evaluation.
- **HuGe** (Liu et al., October 2026) [5] targets the main bottleneck, data. It pre-trains humanoid VLAs on over 2,000 hours of *egocentric* human and robot video. The VLA outputs latent motion tokens for a pre-trained whole-body controller plus hand commands. To bridge the difference between a human and a robot body, it creates pseudo-proprioceptive states by replaying motions in simulation and re-renders the arms in the camera images during fine-tuning. It reports state-of-the-art results on the HumanoidArena benchmark, and on a real Unitree G1 the egocentric data significantly improved success on a pick-and-drop task.

3.4 Comparison

Table 1 summarises the solutions against the criteria in the introduction.

Solution	Architecture	Main strength	Evidence
Atlas (product)	Industrial humanoid; DeepMind models being integrated	Deployment readiness: payload, environment rating, battery self-swap, fleet software	Company specs; first fleets in 2026
GR00T N1	Dual-system VLA (VLM + diffusion action head)	Open weights; real + human-video + synthetic data	Paper; sim and real-robot tests
Gemini Robotics 1.5	VLA + separate embodied-reasoning model	Reasoning before acting; cross-robot motion transfer	Technical report
Helix 02 + Figure 03	Three tiers: S2 reasoning, S1 visuomotor, S0 learned WBC	Single learned system from pixels to whole-body action; long-horizon demo	Company demos only
FAST	Pre-trained WBC + constrained residual adaptation	Fast adaptation without forgetting; CoM-aware balance	Preprint; sim and real
HuGe	VLA $\rightarrow$ latent motion tokens $\rightarrow$ WBC	Scales data with egocentric human video	Preprint; benchmark + G1

Table 1: Comparison of the reviewed humanoid solutions.

4 The best solution and why

In my view the best current solution is the **hierarchical learned stack**: a slow reasoning model (System 2), a fast visuomotor policy (System 1) and a learned whole-body controller (System 0), trained so that the layers share a common interface. The most complete published example is **Figure's Helix 02 on Figure 03**. For a factory buyer today, **Atlas** is the most deployment-ready body, and its pairing with DeepMind's models points to the same architecture. I choose this solution for four reasons.

1. **It matches the physics of the problem.** Balance must be corrected within milliseconds, grasping within tens of milliseconds, and task planning within seconds. One network running at one rate cannot serve all three. The three-tier design gives each timescale its own loop. This is the same principle as the cascaded inner and outer loops we study in control, applied to learned components.
2. **It removes the brittle hand-written boundary between walking and manipulation.** In older humanoids, locomotion and arm control were separate controllers joined by a state machine, which is why many demos showed a robot walking, stopping, and only then reaching. Helix 02's System 0 lets the upper layers command whole-body motion directly, so walking while carrying, bending while reaching, and balancing during contact become one behaviour. The dishwasher demonstration, with dozens of sequential loco-manipulation actions and no resets, is the strongest public evidence of long-horizon whole-body autonomy I found.
3. **It scales with data rather than with engineering effort.** Adding a new task means collecting demonstrations, not writing a new controller. Independent research supports this direction: GR00T N1 and HuGe show that human video and synthetic data can supplement scarce robot data, and the BFM literature shows the WBC layer itself can be pre-trained once and reused.
4. **The research community has converged on it.** GR00T N1's dual-system design, Gemini Robotics 1.5's split between reasoning and action, and HuGe's "VLA outputs tokens for a pre-trained WBC" interface are all variants of the same hierarchy. When independent groups arrive at the same structure, that is good evidence the structure is right, even if the details differ.

I rank Helix 02 first as an integrated system but with a clear qualification: its evidence is company demonstrations, with no public benchmark or paper. If openness and reproducibility are weighted more heavily, GR00T N1 (open weights) and Gemini Robotics 1.5 (detailed technical report) are better references for how the architecture works.

5 Challenges in improving humanoid robots

1. **Data scarcity and embodiment gap.** Language models learned from trillions of words; humanoid policies have thousands of hours of demonstrations. Teleoperation is slow and expensive, and human video does not contain the robot's joint torques or contact forces. HuGe's need for simulation replay and arm re-rendering shows how much work is required just to make human data usable.
2. **Reliability in the long tail.** A demo succeeding once is very different from 99.9% success over an eight-hour shift. Learned policies can fail in surprising ways when lighting, objects or clutter differ from training data, and these failures are hard to predict in advance.
3. **Safety and certification.** A humanoid that falls when power is lost is a new hazard class. Existing standards do not cover it: ISO 10218 does not address mobility, ISO 3691-4

excludes these robots, and ISO 13482 is not for industrial sites. The dedicated standard for dynamically stable industrial mobile robots, ISO 25785-1, is still in development [11]. It is also unclear how to certify a neural network controller whose behaviour cannot be fully specified in advance.

4. **Hands and touch.** Dexterous manipulation still lags far behind locomotion. Multi-fingered hands are fragile and costly, and tactile sensing is only now being integrated into learned policies (for example the palm cameras and tactile sensors of Figure 03).
5. **Energy, compute and cost.** Running large VLA models on board at high rates competes with the battery used for motion. Battery swapping, as on Atlas, works around limited runtime rather than solving it. Unit cost must also fall far enough for a humanoid to beat a fixed automation cell or a human worker on total cost.
6. **Evaluation.** Much of the most impressive work is shown only in company videos. Without shared benchmarks and independent testing, it is difficult to compare systems or know which advance actually caused an improvement.

6 Future directions

- **Scaling data from humans and simulation.** Egocentric human video (as in HuGe), large synthetic datasets, and world models that predict future observations should reduce dependence on teleoperation.
- **Learning from deployment.** Fleets connected through software such as Orbit can pool experience, so a correction learned by one robot improves all of them. Reinforcement learning on real deployment data is a natural next step after imitation learning.
- **Combining learning with model-based guarantees.** FAST's centre-of-mass objectives suggest a hybrid future: learned policies for flexibility, wrapped by model-based safety filters (for example balance or collision constraints solved as a small optimisation) that can be verified and certified.
- **Richer sensing.** Tactile skins, in-hand cameras and force sensing fused with vision, using the estimation tools of this course, will be needed for reliable contact-rich work.
- **Standards and benchmarks.** Completing ISO 25785-1 and creating shared simulation and real-world benchmarks (HumanoidArena is an early example) will move the field from demos to measurable progress.
- **Efficient on-board models.** Smaller, distilled VLAs and dedicated accelerators will cut power and latency so that more of the stack runs locally without a network connection.

7 What I learned from this review

[DRAFT — rewrite in your own words and keep only what is true for you.]

- The classical material in EE6221 has not been replaced by AI; it has moved to a different place in the stack. Kinematics and dynamics now appear as the structure of the whole-body controller, as balance quantities such as the centre of mass fed into a learned policy, and as the retargeting that maps human motion onto a robot's joints. State estimation of the floating base is still a filtering problem.
- The most important design decision in a modern humanoid is how to split the control problem by timescale. This is a familiar control idea (cascaded loops), now applied to neural networks.

- Data, not algorithms, is currently the binding constraint, which is why so much recent research is really about finding new sources of data.
- I learned to separate evidence types: a peer-reviewed paper, a preprint and a company video support claims to very different degrees. [Add one or two specific things that surprised you.]

8 Effect of this assignment on my understanding and interest

[This section must be your own honest answer; the points below are prompts only.]

- **Understanding of robotics and intelligent systems:** [Did it increase? Which link between the lectures (kinematics, dynamics and control, vision sensors, Kalman filtering, vision-based control) and real humanoids became clearer?]
- **Interest in a career in robotics and intelligent systems:** [Did it increase? Which part of the stack (control, perception, learning, safety) would you want to work on, and why?]

References

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